



SUYASH KUMAR

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PERSONAL SUMMARY

I am currently a fourth-year Mechanical Engineering student at IIT (BHU), Varanasi. I have gained hands-on experience in designing complex mechanical structures and utilising various CAD tools for 3D modelling and simulations. At present, I am focusing on the intersection of deep learning and medical imaging, working on Few-Shot Segmentation for MRI and CT scans. This research involves leveraging advanced machine learning techniques to improve the accuracy and efficiency of image segmentation in the medical field, contributing to the development of data-efficient models for biomedical applications.

SKILLS

- Programming Languages: Python, SQL, C, C++, C#
- Machine Learning & AI: PyTorch, TensorFlow, Hugging Face, Langchain, OpenAI APIs, Retrieval-Augmented Generation (RAG)
- Cloud & Data Technologies: MySQL, AWS, Azure
- Development Tools: GitHub, Visual Studio, FastAPI
- Data and analysis: Pandas, Excel, Google Sheets, MATLAB
- Simulation & Design: Ansys Mechanical, Solidworks, Unreal Engine, Unity

EXPERIENCE

- RESEARCH INTERN**, 02/2025 - Current
Indian Statistical Institute Kolkata, Kolkata, India
- Implemented baseline methodologies for performance evaluation and currently researching UNet-based architectural improvements for few-shot learning in medical image segmentation.
 - Experimenting with various deep learning techniques such as Knowledge Distillation (KD) and Contrastive Learning to enhance model performance and generalisation.
 - Investigated the use of Mamba (state space sequence models) for medical segmentation tasks, analysing their effectiveness in comparison to CNN and Transformer-based architectures.

- ACTIVE CLUB MEMBER**, 01/2022 - Current
Masquerades, Varanasi, India
- Scriptwriter for a Mono Act inspired by the lives of Tawaifs (royal courtesans), performed at the 6th Inter-IIT Cultural Festival hosted by IIT Kharagpur.
 - Directed and staged a play inspired by Mahesh Dattani's Dance Like A Man, performed at IIT Kanpur's cultural fest Antaragni and Inter IIT Cultures 7.0, hosted by IIT Patna.
 - Mono-act performer at IIT BHU's cultural fest Kashiyaatra.
 - Mime performer at IIT Delhi's cultural fest Rendezvous.

- SPARK RESEARCH INTERN**, 05/2023 - 07/2023
IIT Roorkee, Roorkee, India
- Conducted research on the design and implementation of dam gates, focusing on identifying and developing effective and efficient construction methods.
 - I attended the 'SEISMIC HAZARD MAPPING OF INDIA' workshop by the International Centre of Excellence for Dams at IIT Roorkee on June 15, 2024.

- FARM VOLUNTEER**, 03/2023 - 04/2023
Jeevantika
- Worked as a farm volunteer and gained experience about organic farming.
 - Emphasized upon growing saffron (kesar) in artificial habitat.

EDUCATION

IIT BHU, Varanasi, Expected in 01/2027
Integrated Dual Degree: Mechanical Department
CGPA: 7.52

Sahu International School, Mahoba, 01/2021
CBSE
95.6%

Maharishi Patanjali Vidya Mandir, Prayagraj, 01/2019
CBSE
94.6%

PROJECTS

- Predictive Modelling of Vestibular Schwannoma Response Using Radiomics and Machine Learning**
- Developed ML models to predict tumor response post-Cyberknife radiosurgery using radiomic features from pre-treatment MRI.
 - Extracted 851 features using PyRadiomics; performed preprocessing (bias correction, normalization) and D3 segmentation with D3 Slicer.
 - Trained and evaluated classifiers (Neural Network, SVM, XGBoost, Random Forest) using nested cross-validation and LASSO-based feature selection.
 - Built a FastAPI-based web service to deploy the prediction pipeline, enabling MRI file upload and automated response prediction.
- Reimagining Online Shopping Experience with Help of Augmented Reality Experience and Machine Learning**
- Reimagined the online shopping experience by creating 3D product models using Unity 3D and integrating augmented reality and machine learning to enhance user engagement and personalize recommendations.
- Ensemble Learning for Robust Medical Image Classification**
- Developed an ensemble of CNN classifiers (e.g., ResNet, EfficientNet, and Mobile Net) to enhance the robustness and accuracy of medical image classification on limited datasets. By combining predictions using majority voting, the model demonstrated improved generalization across diverse medical imaging tasks, such as tumor detection and organ segmentation.

POSITIONS OF RESPONSIBILITY

- Being part of **Technex**, one of Asia’s oldest and India’s largest techno-management fests organised by IIT (BHU), I led end-to-end operations for the 3-day national-scale event — overseeing logistics, vendor management, team coordination, and on-ground execution across 10+ venues and 50+ events. Supervised a team of 50+ volunteers, ensuring smooth execution for 1,500+ registered participants and guests from 500+ institutions nationwide.
- As part of the Inter-IIT Cultural Contingent held at IIT Patna, I directed a stage production of Mahesh Dattani’s *Dance Like a Man*, managing the entire creative and logistical execution. I led a 15-member cast and crew, overseeing rehearsals, set design, lighting, sound coordination, and overall performance quality. The experience required effective leadership, time management, and cross-functional collaboration to deliver a polished production under a competitive and time-bound environment.